COURSE CODE: CIP-B101

LAB TITLE: CIP-

B101_LAB1B_Sytems_Fundamentals_DISK_PARTITION_ANALYSIS

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PLATFORM USED: Windows

INTRODUCTION:

This laboratory practical was designed to introduce the fundamental system and storage concepts required for digital forensic investigation. The lab examined the different layers where digital evidence may exist, including the user, application, operating system, disk, and memory/process layers. It also involved observing programs and processes, identifying process IDs, and understanding their relevance during live forensic examination. Disk structures and partitions were inspected to understand how storage devices are organized and how partition information can be used to calculate disk capacity. In addition, the boot process, partition styles, and file systems were examined to understand how a computer starts and how data is structured and stored. These concepts are important because a digital forensic examiner must understand the underlying system before accurately locating, interpreting, and preserving digital evidence.

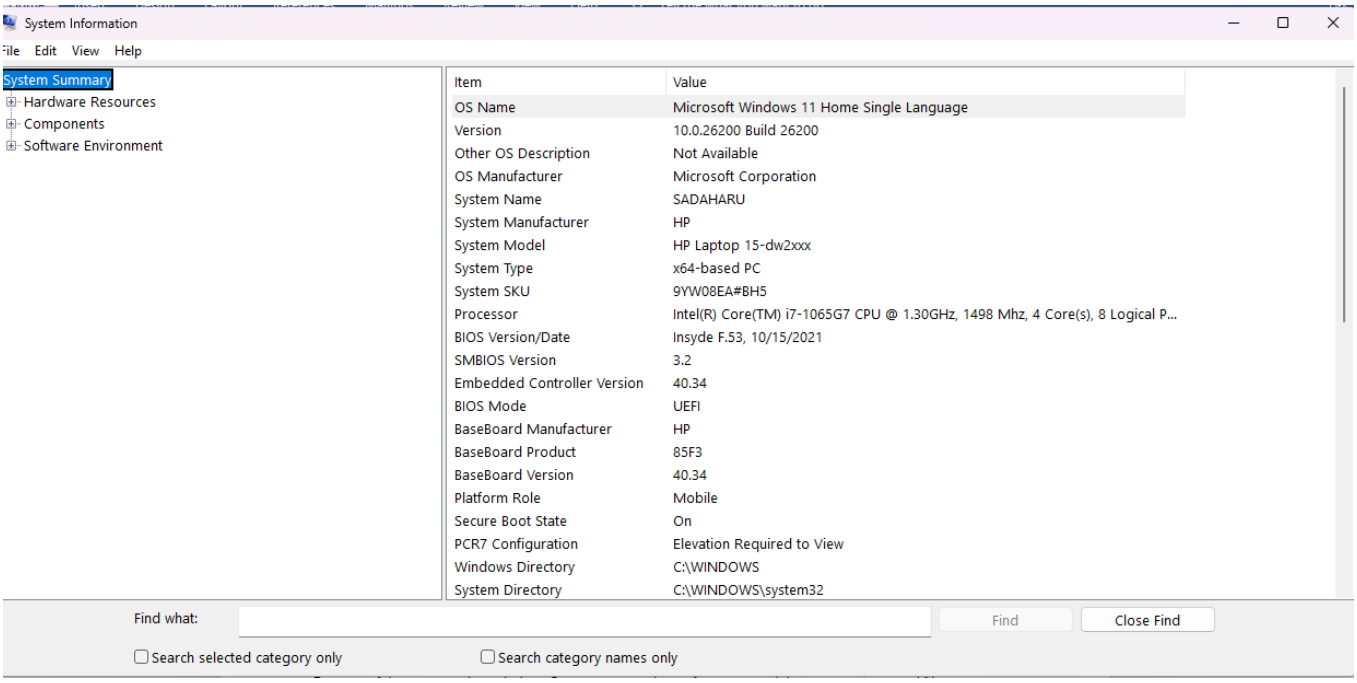
The following tools and environments were used during this laboratory practical:

1. Windows Operating System – used as the primary system environment for system identification and disk inspection.
2. Windows system information utilities – used to obtain information about the computer hardware, firmware, and operating system.
3. Command Prompt and/or PowerShell – used to execute commands and obtain system, process, disk, partition, and file system information.
4. Task Manager – used to observe running processes and their Process IDs (PIDs).
5. Disk Management – used to inspect physical disks, partitions, and storage volumes.
6. Disk and partition commands such as Get-Disk, Get-Partition, Get-Volume were used to verify disk and partition information.

Screenshots were captured throughout the practical to provide evidence of the observations and command outputs obtained from the system.

A - System Identification and Evidence Layer Mapping

The system specification evidence provides important information about the computer being examined, including the operating system, processor, installed memory, firmware configuration, and storage devices. These details are important in digital forensics because they provide context about the environment from which evidence is collected. For example, the firmware mode can help determine whether the system is likely to use BIOS/Legacy booting or UEFI, while the storage configuration can help identify the disk and partition structures that should be examined. Recording system specifications also helps an examiner understand the environment and document the system accurately before conducting further forensic analysis.



Item	Value
Platform Role	Mobile
Secure Boot State	On
PCR7 Configuration	Elevation Required to View
Windows Directory	C:\WINDOWS
System Directory	C:\WINDOWS\system32
Boot Device	\Device\HarddiskVolume5
Locale	United States
Hardware Abstraction Layer	Version = "10.0.26100.1"
User Name	SADAHARU\ABUMA
Time Zone	W. Central Africa Standard Time
Installed Physical Memory (RAM)	12.0 GB
Total Physical Memory	11.8 GB
Available Physical Memory	5.39 GB
Total Virtual Memory	12.5 GB
Available Virtual Memory	5.98 GB
Page File Space	768 MB
Page File	C:\pagefile.sys
Kernel DMA Protection	On
Virtualization-based security	Running
Virtualization-based security Re...	Base Virtualization Support
Virtualization-based security Av...	Base Virtualization Support, Secure Boot, DMA Protection, UEFI Code Readonl...
Virtualization-based security Se...	Hypervisor enforced Code Integrity

Find | Close Find

Search category names only

```
C:\Users\ABUMA>systeminfo

Host Name:                 SADAHARU
OS Name:                   Microsoft Windows 11 Home Single Language
OS Version:                10.0.26200 N/A Build 26200
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:          ABUMA
Registered Organization:   N/A
Product ID:                00327-60000-00000-AA584
Original Install Date:     10/15/2026, 1:12:54 AM
System Boot Time:          11/9/2026, 8:50:07 AM
System Manufacturer:       HP
System Model:              HP Laptop 15-dw2xxx
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 126 Stepping 5 GenuineIntel ~1298 Mhz
BIOS Version:              Insyde F.53, 10/15/2021
Windows Directory:         C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:               \Device\HarddiskVolume5
System Locale:             en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC+01:00) West Central Africa
Total Physical Memory:     12,070 MB
Available Physical Memory: 5,229 MB
Virtual Memory: Max Size:  12,838 MB
Virtual Memory: Available: 5,988 MB
Virtual Memory: In Use:    6,850 MB
```

```
PS C:\WINDOWS\system32> Get-Disk

Number F Serial Number                   HealthStatus         OperationalStat
       r                                                      us
       i
       e
       n
       d
       l
       y

       N
       a
       m
       e
------ - -------------                   ------------         ---------------
0      T            10RIPATKT           Healthy              Online
1      S ACE4_2E00_259F_A570_2EE4_AC00... Healthy              Online
```

```
PS C:\WINDOWS\system32> Get-Partition


   DiskPath: \\?\scsi#disk&ven_ata&prod_toshiba_mq04abf1#4&3b3f12ec&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    100 MB System
2                            105906176                                   16 MB Reserved
3                D           122683392                                 930.26 GB Basic
4                            998979403776                                1.14 GB Recovery


   DiskPath: \\?\scsi#disk&ven_nvme&prod_skhynix_hfs256gd#5&bc33ab1&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    200 MB System
2                            210763776                                   16 MB Reserved
3                C           227540992                                 237.37 GB Basic
4                            255097569280                                917 MB Recovery
```

```
PS C:\WINDOWS\system32> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
                         NTFS           Fixed     Healthy      OK                    60.49 MB   1.14 GB
                         FAT32          Fixed     Healthy      OK                    161.5 MB    196 MB
                         FAT32          Fixed     Healthy      OK                    61.41 MB     96 MB
                         NTFS           Fixed     Healthy      OK                   115.14 MB    917 MB
C                        NTFS           Fixed     Healthy      OK                   162.12 GB 237.37 GB
D                        NTFS           Fixed     Healthy      OK                   617.27 GB 930.26 GB
```

Digital evidence can exist at multiple layers of a computer system. Mapping evidence across these layers helps a forensic examiner understand where particular artefacts may be located and how they can be used to reconstruct user activity or system events. The following table identifies examples of evidence at the user, application, operating system, disk, and memory/process layers and explains their forensic relevance.

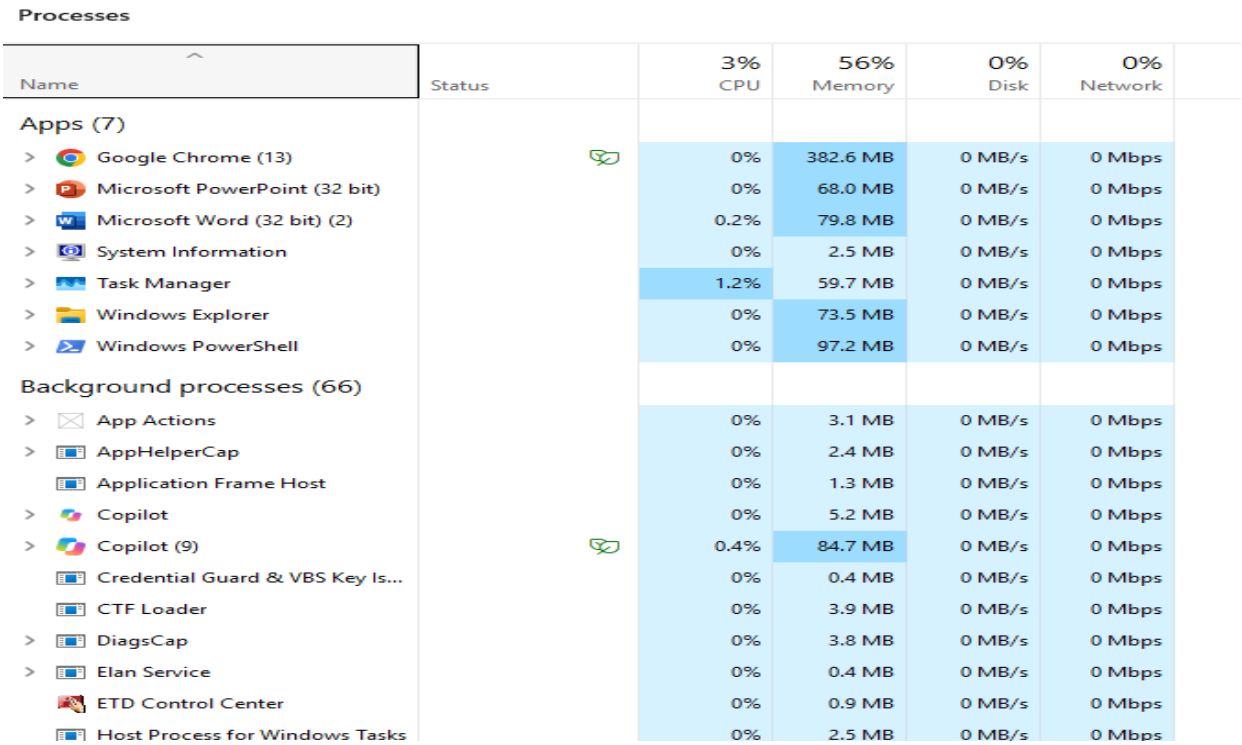
Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
User layer	Opened files, searched terms, typed usernames	Recent files, browser history, shell history	Links user action to system activity
Application layer	Office document metadata, browser cache, chat logs	Application folders and databases	Shows what software created or used evidence
OS layer	Event logs, installed programs, user profiles	Windows logs, registry, Linux /var/log	Shows system events and account activity
Disk layer	Deleted files, file fragments, partitions	File system and unallocated space	Supports recovery and timeline analysis
Memory/process layer	Running malware, PIDs, network connections	RAM capture, process list	Shows live system behavior

B - Program, Process and Thread Investigation

- explain the difference between a program, a process, and a thread in your own words: - A program is a set of instructions stored on a computer that is designed to perform a particular task. When a program is executed, it becomes a process. A process is therefore an active instance of a running program and is assigned resources by the operating system, including memory and a Process ID (PID). A thread is a smaller unit of execution within a process. A single process may contain one or more threads that perform different tasks while sharing the resources of the same process.

- Explain why process information may be useful during malware analysis or live forensic triage: - Process information is useful during malware analysis and live forensic triage because it allows an examiner to identify what programs are currently running on a system. Information such as the process name, PID, parent process, memory usage, and associated network activity can help identify suspicious or malicious behavior. During a live investigation, unusual or unknown processes may provide important evidence of malware execution, unauthorized activity, or compromise. Recording the PID is also important

because it allows the investigator to identify and correlate a specific running process with other forensic evidence




```
PS C:\WINDOWS\system32> Get-Process | Select-Object -First 20 Name,Id,CPU,WorkingSet

Name                     Id      CPU WorkingSet
----                     --      --- ----------
AggregatorHost         6388  0.71875   12369920
ai                     3944   0.5625   36118528
ai                    17060    0.375   37371904
AppActions            15220 0.796875   42450944
AppHelperCap           2424 2.703125   33271808
ApplicationFrameHost   3476 0.359375   49160192
audiodg               11176      0.5   11943936
backgroundTaskHost    12168   0.6875   65728512
chrome                 1536    11.25  106803200
chrome                 2688 0.234375   21032960
chrome                 8172  0.15625   29999104
chrome                 9192 13.21875   51716096
chrome                 9264 1.015625   61386752
chrome                 9864   0.0625   10022912
chrome                10800  0.46875   52883456
chrome                11488 0.890625   20680704
chrome                12868  0.53125   42160128
chrome                14148 48.78125  141475840
chrome                15272       66  230891520
chrome                15632 8.921875   95838208
```

The screenshot shows active processes running on the system and their associated Process IDs (PIDs). The PID uniquely identifies a running process during that particular session and can be useful when correlating process activity with memory, network, or other live forensic evidence.

C - Disk Sector and Disk Size Calculation

Item	Formula / Value	Answer
Total sectors	400 x 12 x 17000	81,600,000 sectors
Total bytes	81,600,000 x 512	41,779,200,000 bytes
Kilobytes	41,779,200,000 / 1024	40,800,000 KB
Megabytes	40,800,000 / 1024	39,843.75 MB
Gigabytes	39,843.75 / 1024	38.91 GB

- Disk and partition inspection was performed to identify how the storage device is organized and to obtain information about the partitions present on the system. Partition information may include the partition number, start sector, end sector or total number of sectors, partition size, and file system type. This information is important in digital forensics because partitions define how storage space is divided and may contain different operating systems, file systems, user data, recovery data, or other forensic artefacts.
- The partition table evidence was used to verify the structure and size of the identified partitions. Command output and screenshots were recorded to ensure that the calculations were based on actual disk information rather than estimated values.

The total disk capacity can be calculated by determining the total number of sectors and multiplying this value by the sector size.

Total sectors:

400 × 12 × 17,000 = 81,600,000 sectors

Total bytes:

81,600,000 × 512 bytes = 41,779,200,000 bytes

Kilobytes:

41,779,200,000 ÷ 1024 = 40,800,000 KB

Megabytes:

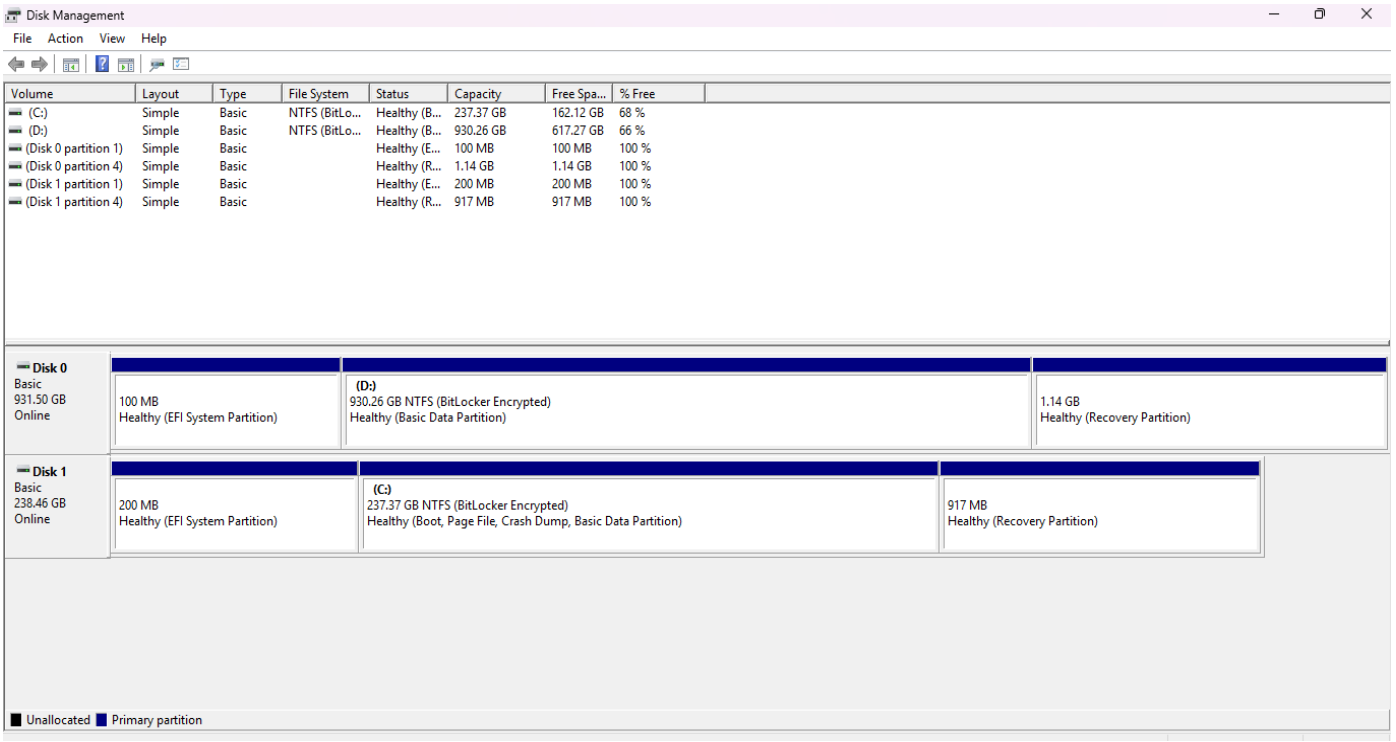
40,800,000 ÷ 1024 = 39,843.75 MB

Gigabytes:

39,843.75 ÷ 1024 = approximately 38.91 GB

Therefore, based on the values provided and the 1024-based conversion used in this calculation, the calculated disk capacity is approximately 38.91 GB.

D - Safe Partition Table Inspection



```
PS C:\WINDOWS\system32> Get-Disk

Number Friendly Name                                  Serial Number                 HealthStatus         OperationalStatus      Total Size Partition
                                                                                                                                      Style
------ -------------                                  -------------                 ------------         -----------------      ---------- ----------
0      TOSHIBA MQ04ABF100                                        10RIPATKT            Healthy              Online                 931.51 GB GPT
1      SKHynix_HFS256GDE9X081N                        ACE4_2E00_259F_A570_2EE4_AC00... Healthy           Online                 238.47 GB GPT


PS C:\WINDOWS\system32> Get-Partition


   DiskPath: \\?\scsi#disk&ven_ata&prod_toshiba_mq04abf1#4&3b3f12ec&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                                   Size Type
---------------  ----------- ------                                                   ---- ----
1                            1048576                                                100 MB System
2                            105906176                                               16 MB Reserved
3                D           122683392                                           930.26 GB Basic
4                            998979403776                                          1.14 GB Recovery


   DiskPath: \\?\scsi#disk&ven_nvme&prod_skhynix_hfs256gd#5&bc33ab1&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                                   Size Type
---------------  ----------- ------                                                   ---- ----
1                            1048576                                                200 MB System
2                            210763776                                               16 MB Reserved
3                C           227540992                                           237.37 GB Basic
4                            255097569280                                           917 MB Recovery


PS C:\WINDOWS\system32> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
                         NTFS           Fixed     Healthy      OK                    60.49 MB   1.14 GB
                         FAT32          Fixed     Healthy      OK                    161.5 MB    196 MB
                         FAT32          Fixed     Healthy      OK                    61.41 MB     96 MB
                         NTFS           Fixed     Healthy      OK                   115.14 MB    917 MB
C                        NTFS           Fixed     Healthy      OK                  162.12 GB 237.37 GB
D                        NTFS           Fixed     Healthy      OK                  617.27 GB 930.26 GB
```

E - MBR, GPT and Boot Process Explanation

Explain the boot process

- Create a simple flow diagram or numbered list explaining how a PC boots.
- Your explanation must include power-on, POST, BIOS/UEFI, CMOS/firmware settings, MBR/GPT or EFI boot records, boot loader/manager, kernel loading, and operating system control.
- State whether your own system uses BIOS/Legacy or UEFI based on system information evidence.

The boot process begins when the computer is powered on. The system performs a Power-On Self-Test (POST) to initialize and check important hardware components. After this, the BIOS or UEFI firmware takes control and applies the configured firmware and boot settings, including identifying the appropriate boot device.

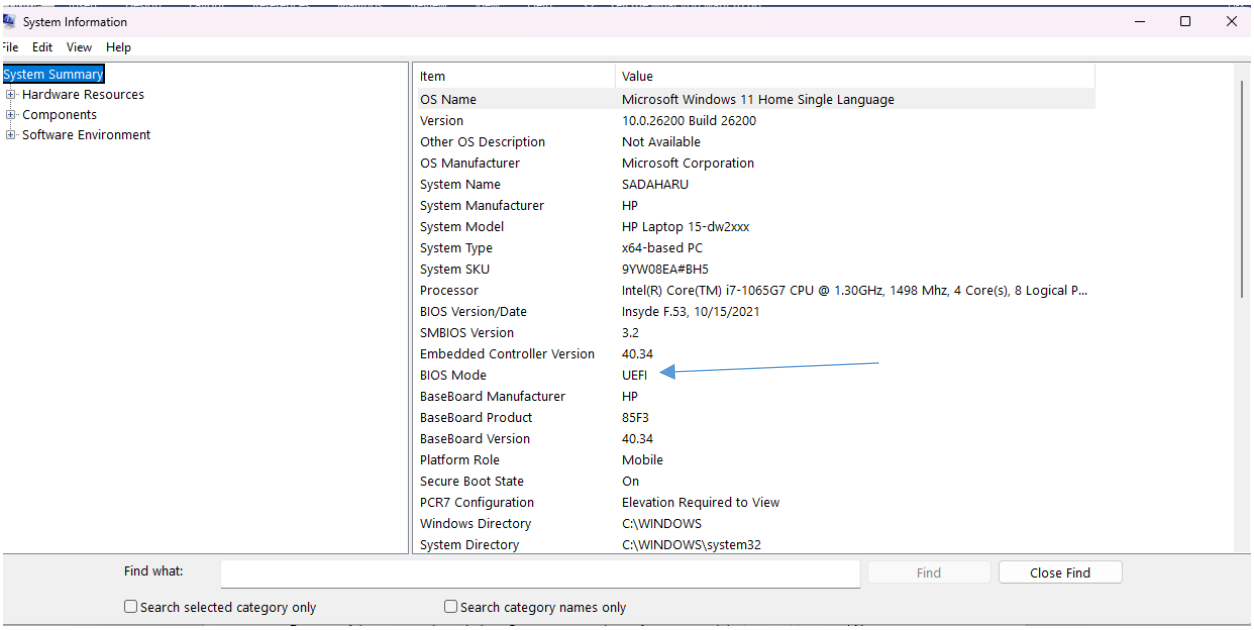
Depending on the system configuration, the computer then locates the relevant boot information. A Legacy BIOS system may use the Master Boot Record (MBR), while a UEFI system typically uses boot information stored through the EFI System Partition and associated boot records. The boot loader or boot manager then identifies and prepares the operating system for loading.

Next, the operating system kernel is loaded into memory and begins initializing the essential components required for the operating system to function. Once this process is completed, control is transferred to the operating system, allowing system services, applications, and user activity to begin.

From a forensic perspective, the boot process is important because boot configuration, firmware settings, boot records, and operating system artefacts may provide evidence about system configuration, tampering, multiple operating systems, or abnormal boot behavior.

1. Power On
↓
2. POST – Hardware Checks
↓
3. BIOS/UEFI Firmware Initialisation
↓
4. Boot Device Selected
↓
5. MBR/GPT/EFI Boot Information Located
↓
6. Boot Loader/Boot Manager Starts
↓
7. Operating System Kernel Loads
↓
8. Operating System Takes Control

Stage	What Happens	Forensic Relevance
Power-on and POST	Hardware is initialized and checked.	Hardware failure or tampering may affect evidence access.
BIOS/UEFI firmware	Boot device order and firmware settings are applied.	Boot mode can explain partition style and boot artifacts.
MBR/GPT or EFI boot	System locates bootable partition or EFI boot entry.	Boot records may contain evidence of boot configuration.
Boot loader/manager	OS selection and pre-kernel loading occurs.	May show multi-boot or hidden OS behavior.
Kernel and OS control	Operating system fully starts.	Logs and user activity begin to accumulate.



Based on the system information evidence, my computer uses UEFI firmware mode. This is consistent with modern systems and is commonly associated with GPT partitioning and EFI boot records.

MBR & GPT:

MBR and GPT are two different methods used to organize partition information on a storage device. MBR is an older partitioning system that stores important boot and partition information in the first sector of the disk. GPT is a newer system commonly associated with UEFI and uses primary and backup partition information for improved reliability. Understanding the partition style is important during forensic analysis because it helps an examiner interpret the disk structure correctly and locate relevant boot and partition artefacts.

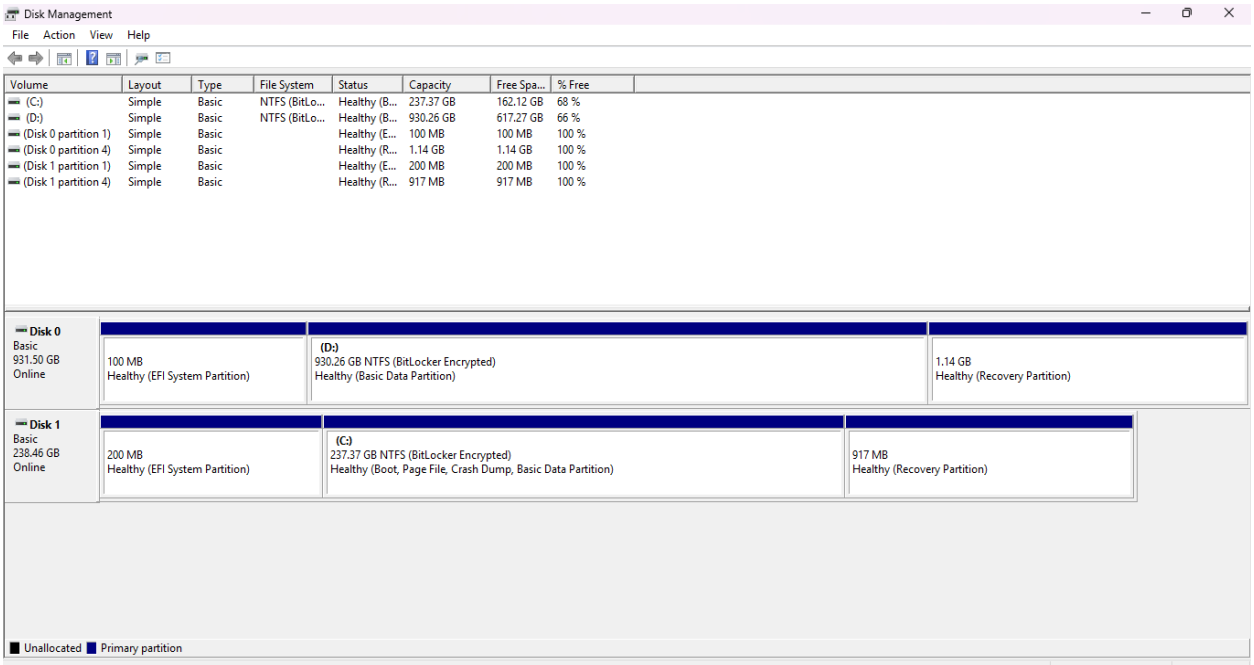
Feature	MBR	GPT
Age / design	Older partitioning system from the 1980s.	Modern partitioning system used with UEFI systems.
Size limitation	Commonly limited to about 2 TB.	Supports very large disks; slide notes up to about 9.4 ZB.
Partition information	Partition table is stored in the first sector with boot code.	Uses primary and backup GPT headers and partition entries.
Forensic issue	Critical information in sector 0; damage can affect partition interpretation.	Backup structures may help recovery and validation.

F - File System Identification and Interpretation

explain why a file system is important in digital forensics

A file system is important in digital forensics because it determines how data is organized, stored, named, accessed, and managed on a storage device. Different file systems store metadata and forensic artefacts in different ways. By identifying the file system used on a partition, an examiner can better understand where to locate files, directories, timestamps, metadata, deleted data, and other evidence.

File system identification is also important when interpreting recovered evidence. For example, the structure and behavior of NTFS may differ from FAT32, exFAT, ext4, or other file systems. Understanding the correct file system helps the examiner select appropriate forensic techniques and avoid incorrectly interpreting disk structures or artefacts.



```
PS C:\WINDOWS\system32> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
                         NTFS           Fixed     Healthy      OK                     60.49 MB   1.14 GB
                         FAT32          Fixed     Healthy      OK                     161.5 MB    196 MB
                         FAT32          Fixed     Healthy      OK                     61.41 MB     96 MB
                         NTFS           Fixed     Healthy      OK                     115.14 MB    917 MB
C                        NTFS           Fixed     Healthy      OK                     162.12 GB 237.37 GB
D                        NTFS           Fixed     Healthy      OK                     617.27 GB 930.26 GB
```

G - Final Practical Assignment

```
PS C:\WINDOWS\system32> Get-Disk | Format-Table Number,FriendlyName,LogicalSectorSize,PhysicalSectorSize,Size -AutoSize

Number FriendlyName           LogicalSectorSize PhysicalSectorSize         Size
------ ------------           ----------------- ------------------         ----
     0 TOSHIBA MQ04ABF100                   512              4096 1000204886016
     1 SKHynix_HFS256GDE9X081N              512              4096  256060514304


PS C:\WINDOWS\system32> Get-Disk 0 | Get-Partition


   DiskPath: \\?\scsi#disk&ven_ata&prod_toshiba_mq04abf1#4&3b3f12ec&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                                                  Size Type
---------------  ----------- ------                                                                  ---- ----
1                            1048576                                                              100 MB System
2                            105906176                                                             16 MB Reserved
3                D           122683392                                                         930.26 GB Basic
4                            998979403776                                                        1.14 GB Recovery


PS C:\WINDOWS\system32> Get-Disk 1 | Get-Partition


   DiskPath: \\?\scsi#disk&ven_nvme&prod_skhynix_hfs256gd#5&bc33ab1&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                                                  Size Type
---------------  ----------- ------                                                                  ---- ----
1                            1048576                                                              200 MB System
2                            210763776                                                             16 MB Reserved
3                C           227540992                                                         237.37 GB Basic
4                            255097569280                                                          917 MB Recovery
```

Disk 0 — Toshiba HDD

- Logical sector size: **512 bytes**
- Physical sector size: **4096 bytes**
- Size: **1,000,204,886,016 bytes** (~1 TB)

Disk 1 — SK hynix SSD

- Logical sector size: **512 bytes**
- Physical sector size: **4096 bytes**
- Size: **256,060,514,304 bytes** (~256 GB)

Partition 1 – Disk 0 (TOSHIBA MQ04ABF100)

The logical sector size of the disk was identified using the Get-Disk command as 512 bytes.

Partition 1 Offset = 1,048,576 bytes

Start Sector = Partition Offset ÷ Logical Sector Size = 1,048,576 ÷ 512 = 2,048

Therefore, Partition 1 starts at **sector 2,048.**

The next partition begins at an offset of **105,906,176 bytes: 105,906,176 ÷ 512 = 206,848**

Therefore, Partition 1 ends immediately before **sector 206,848**.

Number of sectors in Partition 1:

206,848 – 2,048 = 204,800 sectors

Partition size: **204,800 × 512 = 104,857,600 bytes**

104,857,600 ÷ 1,073,741,824 = approximately 0.0977 GB

Therefore, Partition 1 has a calculated size of approximately 0.0977 GB (approximately 100 MB), which corresponds with the System partition size reported by Windows.

Partition 2 – Disk 0 (TOSHIBA MQ04ABF100)

Partition 2 Offset = 105,906,176 bytes

Start Sector = 105,906,176 ÷ 512 = 206,848

Therefore, Partition 2 starts at **sector 206,848.**

The next partition begins at an **offset of 122,683,392 bytes: 122,683,392 ÷ 512 = 239,616**

Therefore, Partition 2 ends immediately before sector 239,616.

Number of sectors in Partition 2: 239,616 – 206,848 = 32,768 sectors

Partition size:

32,768 × 512 = 16,777,216 bytes

16,777,216 ÷ 1,073,741,824 = approximately 0.0156 GB

Therefore, Partition 2 has a calculated size of approximately 0.0156 GB (approximately 16 MiB), which corresponds with the Microsoft Reserved partition size reported by Windows.

Main assignment

Verify the size of partitions 1 and 2 of a PC using a partition table. Record the verification process step-by-step. Calculate the size of each partition in GB. No full credit will be awarded if you do not use partition table evidence.

Task	Required Evidence	Student Notes / Answer
Identify partition 1	Screenshot of partition table and command output	

Record partition 1 sector count	Start sector, end sector or sector count	
Calculate partition 1 size	Formula and result in GB	
Identify partition 2	Screenshot of partition table and command output	
Record partition 2 sector count	Start sector, end sector or sector count	
Calculate partition 2 size	Formula and result in GB	
Compare with OS-reported size	Disk Management, Get-Volume, lsblk, or df output	
Explain difference if any	Short explanation of rounding, reserved space, file system overhead, or GiB/GB differences	

Conclusion:

This laboratory practical provided a practical understanding of important concepts required for digital forensic investigation. The exercise demonstrated how digital evidence can exist across different layers of a computer system, including the user, application, operating system, disk, and memory/process layers. It also demonstrated the importance of examining running processes and Process IDs during live system analysis. Disk and partition inspection showed how storage devices are organized and how partition sizes can be verified using sector information and manual calculations.

The practical also improved understanding of the computer boot process, the differences between MBR and GPT partition structures, and the importance of identifying the correct file system. These concepts are important in digital forensics because accurate interpretation of the underlying system helps an examiner locate, analyze, and correctly interpret digital evidence. Overall, the lab demonstrated that effective forensic analysis requires both technical knowledge of computer systems and careful documentation of observations and evidence.